

Theoretical Questions

Question 1 : What is Data Loading in Power BI and why is it considered the first step of analysis?

Data Loading in Power BI is the process of importing data from different sources into Power BI Desktop so it can be analyzed and visualized.

Data sources can include: Excel files , CSV files , Databases (SQL Server, MySQL) , Web data , Cloud services

Data loading is considered the first step because:

1. Power BI cannot perform analysis without data.
2. It allows users to connect to multiple data sources.
3. It brings the raw data into Power BI for further cleaning and transformation.
4. It prepares the data for modeling and visualization.

Example:

Loading the Corona_Virus_2020 dataset into Power BI before cleaning duplicates or calculating recovery rate.

Question 2 : Explain the difference between “Load” and “Transform Data” in Power BI.

Load	Transform Data
Directly imports data into Power BI model.	Opens Power Query Editor to clean and modify data before loading.
No changes are made to the data.	Data can be cleaned, filtered, or reshaped.
Faster when data is already clean.	Used when data requires preparation.
Done by clicking Load in Power BI.	Done by clicking Transform Data.

Question 3 : What is a Fact Table and a Dimension Table? Give examples from the dataset.

Fact Table

A Fact Table contains numerical data or measurable values used for analysis.

Characteristics:

- Contains metrics and calculations
- Usually large in size
- Connected to dimension tables

Examples from the dataset: Confirmed_Cases , Recovered_Cases ,Deaths , Active_Cases , Tests_Conducted

These represent measurable COVID-19 statistics.

Dimension Table

A Dimension Table contains descriptive information used to categorize or filter data.

Characteristics:

- Contains text or descriptive attributes
- Used for filtering and grouping
- Usually smaller in size

Examples from the dataset: Country , State , Date ,Vaccination_Status

These help analyze facts like cases by country or date.

Question 4 : Why is Star Schema preferred over Snowflake Schema in Power BI?

Star Schema is preferred in Power BI because it is simpler and faster for reporting and analysis.

Reasons:

1. **Better Performance** - Fewer table joins compared to Snowflake Schema. Faster query execution.
2. **Simpler Data Model** - Easy to understand for users and developers.
3. **Efficient Visualization** - Power BI works better with a central fact table connected to dimension tables.
4. **Improved Data Relationships** - Easier to manage relationships between tables.

Example (Star Schema for the dataset)

Fact Table

- **COVID_Statistics**
 - Confirmed_Cases
 - Recovered_Cases
 - Deaths

Dimension Tables

- Country
- State
- Date
- Vaccination_Status

Practical Questions

Question 5 : Identify and remove duplicate records based on Date, Country, and State.

Question 6 : Identify and replace null values in

Question 7 : Create a new column to calculate Recovery Rate.

Link - [Q 5,6,7](#)

Question 8 : Create a summarized table showing total confirmed cases by Country. Corona_Virus_2020
Vaccination_Status.

Country	Sum of Confirmed_Cases
Brazil	4804531
India	4198680
Italy	3390760
Spain	4354613
UK	3651598
USA	3399318
Total	23799500